

Pharmacotherapy of Gout and Pseudogout

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Slides 1 and 2

Hi, everyone. This is Dr. Dosha Cummins. I am a faculty member on the Arkansas campus. I will be reviewing the pharmacologic management of gout. My email address is on this first slide as well as my work phone number. Please feel free to reach out to me if you have any questions about this topic. I am more than glad to discuss any questions you have via Zoom or email or via phone. Objectives were outlined for review on the CPG.

Slide 3

Regarding the pharmacologic management of gout, we are considering two broad categories of management. One is the treatment of an acute episode or an acute attack, which includes drug therapies that target inflammation. We primarily use nonsteroidal agents, glucocorticosteroids, or colchicine. One of these is not necessarily preferred over another, but there are preferences when considering a patient's co-morbidities. We will discuss less commonly used agents for acute gout under 'other.'

I've placed an asterisk by NSAIDS, corticosteroids, and colchicine. This is to reflect that these agents can also be used for the acute management of pseudogout, or calcium pyrophosphate disease. Beyond that, I will not be speaking further about pseudogout. We focus on the management of gout.

The second bullet point is the second broad category in the treatment of gout, which is the prophylaxis of future episodes (after an acute episode) in patients whom we identify prophylaxis is necessary. We use urate-lowering therapy (ULT) in those patients. We again have a limited number of options, three medications. Two of those are xanthine oxidase inhibitors and one is the drug probenecid.

Slide 4

The next slide repeats what I just showed you except I have added between the two broad categories of treatment the transition to ULT. This is for patients in whom you think it is appropriate to start prophylactic therapy following an acute episode. There is a transition process when ULT is started after the acute episode is under control. The patient is on an anti-inflammatory agent plus a ULT for a period of time.

I will reiterate this point later in this discussion. Patients should remain on an anti-inflammatory agent when ULT is initiated because starting ULT can predispose the patient to having another acute gout flare. This is due to uric acid shift/redistribution when ULT agents are started, predisposing these patients to another flare. We do not want to do that when we have just finished or started treating an acute flare. So, there is a need for a transition.

Slide 5

Focusing on acute treatment when a patient comes in with an acute episode:

General principles include treating acute episodes with pharmacotherapy. From a chronic disease or management standpoint, there are non-pharmacologic management considerations such as dietary adjustments. However, in an acute episode, we treat with drug therapy because the episode is extremely painful. Since we are managing an inflammatory process, it is ideal to start treatment within the first 24 hours. Patients are going to want to do this because of the pain. Beyond 24 hours, those inflammatory mediators are well established, and it is more difficult to manage.

Key point: A patient who has an acute flare while taking ULT (previously started to prevent attacks) should continue their ULT if they are experiencing an acute episode. If the ULT is stopped, it will have to be re-started at some point, so continue it during an acute episode. The anti-inflammatory agent of choice will treat the acute episode; continue the ULT. This is a patient counseling point to cover with patients when ULT is prescribed.

Slide 6

Again, there are three categories of medications primarily used for managing gout. One is the non-steroidal agents NSAIDS. You will learn in more detail about NSAIDS elsewhere in the curriculum. I will cover the highlights here.

NSAIDS inhibit prostaglandin synthesis, which is why we are utilizing them in gout - to knock down inflammatory mediators; there is an analgesic response, as well. In addition to being used for acute gout management, NSAIDS can be used in that transition to ULT; the background anti-inflammatory mechanism is desired as ULT is started.

NSAID toxicities can be numerous. I've highlighted some of the more common ones, dyspepsia, GI ulceration, potentially renal toxicity, and also bleeding because of that plate inhibitory effect. There are three NSAIDS that are FDA-approved for managing gout. However, we can use any of the nonsteroidals for treating gout. Some nonsteroidals must be dosed at higher doses to achieve an anti-inflammatory effect versus the analgesic effect achieved at a lower dose. The example I like to use is ibuprofen. For instance, when you have a headache and you take 400 milligrams; for an anti-inflammatory effect, you generally need more than that, such as 600 mg or 800 mg (per dose). That's not something you have to memorize from a dose standpoint, but I do want you to be aware of that concept because it's important when we are treating a condition such as gout.

Lastly, this is a drug class (NSAID) that may not be the best choice in patients with baseline renal insufficiency. Also, elderly patients may not tolerate NSAIDS toxicities because of comorbidities.

Slide 7

A second anti-inflammatory class used in gout is the corticosteroids. They are primarily used just for acute management. Steroids are not typically used for prophylaxis because they do not impact (lower) uric acid levels; they are rarely used for the transition period to ULT because of their toxicity profile.

There are numerous potential adverse effects with corticosteroids. Short-term use side effects include insomnia, mood changes, hyperglycemia, and fluid retention. However, for some patients a steroid may be the best choice for an acute flare. Specific dosing is included on the slide, but you are not expected to memorize the dosing. I do want you to be cognizant of the fact that when we do use prednisone for acute gout, it is important to use a sufficient dose. Underdosing may not provide enough benefit to treat the flare and provide sufficient relief.

Oral prednisone is often as the corticosteroid for acute gout. You may administer an intra-articular injection you have a patient who cannot tolerate oral steroids. Lastly, you may have an occasion where an IM steroid is the best choice for the patient.

Slide 8

The last anti-inflammatory 'class,' which only one medication within that class for acute gout is the drug colchicine. Colchicine has several mechanisms of action, including inhibiting microtubular polymerization. It can be very effective in the management of gout. It is used outside of just acute treatment. It may be used again in the ULT transition period. It can also be used for long-term prophylaxis, but it is used less frequently for this.

We are typically referring to medications that lower uric acid levels when we talk about prophylaxis of gout. Again, colchicine does not impact uric acid levels. It is primarily used in acute gout for its anti-inflammatory effects, but it may be used for prophylaxis when patients cannot tolerate the true uric acid lowering therapies.

Toxicities limit colchicine's use; they include GI effects (nausea, vomiting, diarrhea). Neuropathy and bone marrow suppression are associated with colchicine, but this is primarily with long-term use, which is one of the reasons we do not see it used for prophylaxis.

Colchicine interacts with numerous medications; there are too many to list, but please remember that this is a consideration if a patient is on concomitant medications, particularly those that inhibit the CYP3A4 system. You may have to adjust the colchicine dose when treating an acute episode. There are also interactions via P-gp, as well. Thus, colchicine can be a complicated drug to use, but we are using it on a short-term basis for the acute treatment.

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Colchicine works best if initiated within 36 hours of symptom onset. Typically, patients will start a medication or request treatment well before 36 hours but be aware from a pharmacologic standpoint that the drug does not work as well after 36 hours. This is because that inflammatory process is very well established.

I do want you to know this one dose for this lecture because it's very specific. The dose for acute treatment of gout is 1.2 milligrams once (2 x 0.6 mg); an hour later the patient takes another tablet (0.6 mg). The patients two 0.6 tablets and then an hour later they take one table, and that is it. That is the entire dosing for acute gout. St one time, a much different dosing strategy was used, which made patients very sick from a GI toxicity standpoint.

Patients who continue colchicine for prophylactic therapy are a smaller once-daily dose.

Slide 10

The last category/group of treatments for acute gout is reserved for those patients who do not respond to NSAIDS, corticosteroids or colchicine (or have contraindications). This group the biologics. Humans can no longer make a functional uricase enzyme due to genetic mutations; we cannot break down uric acid. There is a synthetic version of uricase, which is an option for patients who do not respond to other therapies.

There is also an IL-1 inhibitor specifically indicated for gout, which is another option when other acute treatment options have not benefited the patient or there are contraindications. Lastly, a synthetic version of ACTH is available.

Slide 11

Now we will discuss gout prophylaxis which, as mentioned, typically involves ULT.

Does every patient need ULT? No. There is guidance from the American College of Rheumatology: a patient who has more than one tophus, radiological joint damage due to gout, or if a history of two or more gout attacks in a year. These patients will benefit from ULT.

To reiterate the earlier point regarding transitioning to ULT: ULT may be started following an acute if an anti-inflammatory agent has already been initiated. Again, the rationale for both is because ULT 'mobilizes' uric acid into different places and an acute attack or flare can be precipitated. The 'overlap in therapy (anti-inflammatory + ULT) is the transition period.

The goal with ULT is to reduce baseline uric acid levels to < 6 mg/dL.

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So how do we lower uric acid? There are two drug classes: xanthine oxidase inhibitors are one class. Hypoxanthine is a breakdown product of purine metabolism. It is normally eliminated via xanthine oxidase in a couple of steps to become uric acid. Allopurinol, which

is a purine itself, competitively inhibits xanthine oxidase. Thus, less uric acid is formed via the normal pathway because allopurinol is inhibiting the enzyme that ultimately is responsible for the formation of uric acid.

Note: Although most patients under-excrete uric acid (vs over-produce), xanthine oxidase inhibitors are commonly prescribed due to ease of use and efficacy.

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Allopurinol's mechanism of action is summarized in the first bullet. It is currently our preferred first line ULT for the chronic management of gout. It is also indicated for the treatment of hyperuricemia associated with cancer treatment to prevent tumor lysis syndrome.

Its metabolite, oxypurinol (also known as alloxanthine), also inhibits xanthine oxidase. It's a very effective drug therapy. Adverse effects that are mild and reversible include dyspepsia, headache, and diarrhea. A critical point to remember, particularly since so many patients take allopurinol, is that patients should immediately stop the medication if they develop a rash. There is a very specific hypersensitivity associated with the HLA-B*5801 allele. These patients can have a pronounced hypersensitivity that involves rash, eosinophilia, leukocytosis, hepatitis, and renal failure. Counsel patients taking allopurinol to stop taking it if a rash develops and to seek medical attention. It is possible for patients to develop a rash that's not related to this hypersensitivity, but they need to seek medical attention for further evaluation for other findings (eosinophilia, etc.)

Appropriate allopurinol dosing is important, and this is something I do want you to know. Patients should be initiated on a low dose and titrated up to the clinical endpoint, such as goal serum uric acid level. Titrating the dose will help minimize adverse effects, as well as reduce risks in patients with the HLA-B*5801 allele. Note that we do not routinely screen patients for this allele.

Allopurinol can be used in patients with chronic kidney disease; these patients often have hyperuricemia. However, the dose needs to be adjusted based on renal function.

Please note the drug interactions with allopurinol: well-known ones include other drugs that utilize xanthine oxidase, such as azathioprine and mercaptopurine, which are purine derivatives; if they cannot be metabolized, there is an increased risk of adverse effects.

Liver function tests (LFTs) should be monitored with allopurinol, as well as serum creatinine, and CBC (complete blood count).

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Another xanthine oxidase inhibitor is febuxostat. It does not have a purine structure, so it itself is not a substrate for the enzyme. It is only indicated for the chronic management of gout.

Febuxostat is not renally eliminated, but is hepatically eliminated. Thus, dosing is less complicated than with allopurinol, particularly for patients with chronic kidney disease.

It has some similar toxicities as allopurinol such as GI (nausea); LFTs should be monitored. Febuxostat does have a warning about a potential increased risk of cardiovascular death in patients who have a history of vascular disease, which is a factor in why it is not considered a first-line agent over allopurinol.

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Since it inhibits an oxidase, febuxostat shares Dis with allopurinol (azathioprine and mercaptopurine). Similar monitoring is also recommended (LFTs, CBC). You do not need to memorize baseline and every two to four months specifically. I do want to point out that with newer therapies such as febuxostat, sometimes that monitoring is very specific. You can always look that up. Serum creatine monitoring is not needed because it is hepatically eliminated, which is a distinction between it and allopurinol. I have included the initial dose; febuxostat dosing must also be titrated, but the overall dosing range for is narrower than with allopurinol.

Slide 17-18

The final ULT/prophylactic drug to review is the uricosuric class; we only have one currently in use, which is probenecid.

As reminder, uric acid elimination involves glomerular filtration, secretion, and reabsorption. Reabsorption is the primary method of how the kidney manages uric acid. If we look at specific transporters, there are several, but the primary transporter for uric acid in the proximal tubule is URAT1. Inhibiting URAT1 reduces the reabsorption of uric acid. The result is an increased amount of uric acid in the urine. Patients on probenecid must drink sufficient fluids because they are now predisposed to developing uric acid stones. If a patient has a history of uric acid stones (urolithiasis), probenecid is not the best drug for them. Lastly, probenecid should be avoided in patients who are uric acid over-producers.

See bullet point on Slide 12

When we look at the chronic management of gout, many patients take allopurinol. However, most gout patients not over producers of uric acid; they are uric acid under excreters (because human no longer have functional uricase). We still use a medication that inhibits the formation of uric acid, even though most people are not overproducers, simply because a medication such as allopurinol is effective in these patients and relatively easy to use compared to probenecid.

As you might guess, other substances (medications) are impacted by or impact the uric acid transporters. Thus, there are some drug interaction considerations with probenecid.

Note that probenecid is not effective in patients who have compromised renal function (eGFR < 60 mL/min) because it cannot get to where it needs to work.

Slide 19

Back to that transition period from acute treatment to prophylactic treatment when patients are on an anti-inflammatory drug + ULT:

Patients typically are on an anti-inflammatory plus our ULT for three to six months. After this transition period, the anti-inflammatory agent is stopped and the ULT medication is continued for long-term prophylaxis.

Slide 20

In summary, the pharmacologic management of acute gout episodes should be individualized based on the comorbidities of the patient. If a patient comes to you with an acute episode and they are already on ULT, continue the ULT.

If the patient is on an anti-inflammatory agent for an acute episode and they have not previously been on a ULT therapy, one can be initiated if prophylactic therapy is warranted.

Here are the core references for the lecture. Please reach out if you have questions. Thank you.